

Operator ETL — Executive One-Pager

****Agentic data intake for FOIA and public comments****

The problem

Government agencies intake public comments and FOIA requests. Chatbots with database access fail on PII leakage, hallucinated counts, and non-auditable runs.

The answer: three planes

Data	Deterministic bronze → silver → gold ETL (DuckDB local, BigQuery)
Policy	PII scan, encrypted vault, fail-closed before any insight
Control	LangGraph orchestration, MCP allowlisted tools, critic verifies every number

****Core claim:**** Agents orchestrate. Python and SQL execute. Every insight number must exist in gold metrics.

FOIA MVP (proven)

`./harness/e2e.sh` # OKF + 24 tests + FOIA demo

Sample comments	12 (EPA/FCC dockets)
Silver (valid)	10
Quarantined	2
PII flagged	4+
Critic	Pass — no hallucinated counts

Agentic best practices

- MCP allowlist — no raw SQL, no vault decrypt
- LangGraph checkpoints — resumable runs
- Quality gate — KPIs withheld when quarantine rate exceeds threshold
- Structured audit — ``pipeline_runs`` + graph state

What's implemented vs next

Now (IMPLEMENTED)	Next (PARTIAL / SPECIFIED)
Local DuckDB pipeline	BigQuery gold dialect lift
LangGraph + PII + critic	HITL officer dashboard
MCP stdio + HTTP scaffold	Regulations.gov adapter
GCP Terraform + Cloud Run	Production hardening

For adopters

1. ****Prove:**** ``./harness/e2e.sh``
2. ****Run:**** ``uv run etl-graph --source public_comments``
3. ****Deploy:**** ``infra/terraform`` + ``cloudbuild.yaml``

Full engineering spec: `*Operator-ETL-White-Paper.pdf*`

****August 2026**** · Operator ETL · Private repository — share PDFs only